IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 6 February 2025, accepted 4 March 2025. Date of publication 00 xxxx 0000, date of current version 00 xxxx 0000.

*Digital Object Identifier 10.1109/ACCESS.2025.3550057*

# Utilizing Conditional GANs for Synthesis of Equilibrated Hyperspectral Data to Enhance Classification and Mitigate Majority Class Bias

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**ABSTRACT** A land cover scene is described by vegetation, urban infrastructure, water bodies, bare soil, and many other physical materials. Satellite imagery and remote sensing techniques analyze the scene for the management of natural resources and environmental changes. Hyperspectral Imaging (HSI) plays a vital role in land cover classification due to its rich spectral information, which captures a broader range of wavelengths. This enhanced spectral data allows for more precise identification and differentiation of land cover types. Imbalanced class distributions in HSI classification hinder rare class performance, as these critical classes are often underrepresented in training data. GANs offer a solution by generating synthetic samples to address this scarcity. However, current GAN-based methods use a static up-sampling mechanism for minority classes and generate samples for even larger classes. Furthermore, these models use the largest class as the scaling factor for minority classes, leading to excessive synthetic sample generation for smaller classes. Although these methods achieve class balance, but often suffer from synthetic bias due to over-reliance on generated samples, especially for minority classes. As a result, this can cause dataset inflation, compromise the authenticity of minority class representations, and potentially degrade the model's performance on real-world data. To overcome this limitation, we introduce a novel approach that dynamically balances class distributions by leveraging the smallest class size as a scaling factor. Our proposed model, Spectra Balance GAN (SB-GAN), ensures efficient and adaptive sample generation, producing only the necessary synthetic samples while preserving the integrity of the dataset and maintaining a manageable size. The primary novelty of this model lies in its adaptive nature of class balancing mechanism and an integrated conditional GAN to generate only the required samples, ensuring efficient and balanced hyperspectral image classification. The efficacy of the SB-GAN approach has been evaluated using three publicly available datasets: Indian Pines, Pavia University, and Salinas. The approach achieved an average classification accuracy of 99% across these datasets, demonstrating its potential to significantly improve the classification of rare classes in HSI scenes.

**INDEX TERMS** Hyperspectral imaging, HSI, GAN, cGAN, generative adversarial network, conditional generative adversarial network, down sampling, up sampling, synthetic, augmentation, imbalance class, land cover scene.

The associate editor coordinating the review of this manuscript and approving it for publication was Domenico Rosaci.

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## I. INTRODUCTION

Land Cover Scenes (LCSs) refer to the analysis and classification of various physical features on the Earth's surface, such as forests, water bodies, urban areas, and agricultural fields. These scenes are captured using satellite imagery and aerial photography. This remote sensing classifies LCSs into different types based on their characteristics. The classification is crucial for environmental monitoring, urban planning, agricultural development, forestry conservation, and disaster management. Continuous monitoring of LCSs over time identifies deforestation or urban expansion for real time intervention. The specialized remote sensing software processes the data of Geographic Information Systems (GIS) for reasonable classification and interpretation. LCSs classification plays a pivotal role in various applications such as environmental monitoring, urban planning, and agricultural management. Accurate classification of LCSs from remote sensing data is essential for informed decision-making in these fields.

Hyperspectral Imaging (HSI) is a powerfull data in remote sensing and image analysis, capturing information across hundreds of narrow and contiguous spectral bands. Unlike traditional imaging techniques that capture data in a few broad bands (e.g., RGB), HSI provides detailed spectral information for each pixel in an image, offering insights into material composition, vegetation health, and environmental conditio9ns. Each pixel in an HSI dataset corresponds to a spectrum, which can be analyzed to identify unique spectral signatures characteristic of different materials or land cover types. However, HSI data comes with challenges such as high dimensionality, requiring sophisticated processing techniques to handle mixed pixels and ensure accurate classification. Despite these complexities, HSI continues to be invaluable in fields like agriculture, environmental monitoring, mineral exploration, and urban planning, where detailed spectral information is critical for decision-making and analysis.

HSI classification involves several challenges such as limited labeled samples, high dimensionality, and the complexity of handling mixed pixels, which are crucial for ensuring robust model generalization and transferability. Addressing the scarcity of labeled samples is critical for improving classification accuracy, often tackled through generative models that augment existing labeled data. Deep learning models are particularly effective in this context, facilitating the augmentation process to enhance the performance of HSI classification models. However, the performance of deep learning models is often compromised by the inherent class imbalance present in land cover datasets. This imbalance arises when certain land cover types, such as urban areas or water bodies, are underrepresented compared to others like forests or agricultural fields. The disproportionate representation of classes can lead to biased models that underperform in identifying minority classes, thereby affecting the overall accuracy and reliability of the classification system.

Imbalanced datasets pose significant challenges to traditional classification algorithms, which tend to be biased towards the majority class, leading to poor predictive performance for the minority class. This bias is particularly problematic in land cover classification where accurate detection of minority classes can be crucial for tasks such as detecting rare habitats or monitoring small-scale environmental changes [10].

In HSI classification using deep learning, addressing class imbalance is crucial for accurate and robust model performance. Majority undersampling involves randomly reducing instances from the majority class to balance its representation with the minority class, mitigating biases towards overrepresented classes. Conversely, minority oversampling techniques generate synthetic instances or duplicate existing ones from the minority class to increase its sample size, ensuring the deep learning model learns from both classes effectively. These approaches are essential in HSI classification tasks where certain land cover types or materials may be rare but significant for applications such as environmental monitoring or mineral exploration. Implementing these techniques requires careful consideration of model architecture, training procedures, and evaluation metrics to prevent overfitting and optimize classification accuracy across diverse spectral bands and environmental conditions. Addressing class imbalance problems using oversampling the minority class or undersampling the majority class, in their naive form has limitations. Oversampling can lead to overfitting, while undersampling may result in the loss of valuable information from the majority class. Oversampling is often preferred over undersampling for handling imbalanced datasets because it retains all available data, improving the model's ability to learn from the minority class without losing potentially valuable information. Additionally, oversampling maintains the original distribution of the majority class, which can lead to more robust models and better generalization to new data. Conversely, undersampling can distort the data distribution and result in the loss of important information from the majority class.

Generative models enhance the representation of underrepresented classes by generating synthetic data, effectively balancing class distributions. The necessary samples for each class are generated to avoid excessive augmentation and mitigate overfitting risks associated with oversampling. The use of Generative Adversarial Networks (GANs) [3] proves beneficial in this context. It facilitates the generation of realistic synthetic data crucial for improving model performance without compromising generalization.

GANs have emerged as a powerful tool for generating synthetic data. GANs are a class of machine learning frameworks designed by Ian Goodfellow and his colleagues in 2014. GANs consist of two neural networks, the generator and the discriminator, which are trained simultaneously through adversarial processes. The generator creates synthetic data samples, while the discriminator evaluates them alongside real data, distinguishing between genuine and fake samples. The generator aims to produce increasingly realistic

data to fool the discriminator, which in turn, improves its ability to detect fakes. This dynamic, competitive training process leads to the generator producing highly realistic data samples. GANs have shown remarkable success in various applications, including image generation, video synthesis, and data augmentation, pushing the boundaries of what is achievable in artificial intelligence and machine learning. By leveraging the capabilities of GANs, it is possible to create high-quality synthetic instances of HSI that can be used to balance imbalanced datasets.

Conditional Generative Adversarial Networks (cGANs) [8] are an extension of GANs that incorporate conditional information into the model. This additional information can be any type of auxiliary data, such as class labels or data from other modalities. By conditioning both the generator and discriminator on this extra information, cGANs allow for a more controlled and targeted generation of synthetic data. This results in the ability to produce specific types of outputs based on the provided conditions. cGANs have been effectively used in applications such as image-to-image translation, text-to-image synthesis, and super-resolution, enabling more precise and context-aware data generation. cGANs by generating data conditioned on specific class labels, making them particularly suitable for addressing class imbalance in classification tasks. By generating synthetic samples for the minority classes, cGANs can effectively augment the training dataset, thus mitigating the imbalance and enhancing the performance of classification algorithms.

Current GAN-based methods generate rare class samples but often overlook the importance of maintaining a balanced and representative dataset by dynamically adapting to class distributions. These models tend to exhibit synthetic bias, as they over-rely on generated samples, particularly for minority classes. Consequently, this can lead to dataset inflation, reduced authenticity of the minority class representations, and potential degradation of model performance on real-world data.

This research focuses on the application of cGANs to generate synthetic instances for rare classes in LCSs classification tasks. The primary objective of this study is to mitigate synthetic bias through the implementation of an adaptive up-sampling technique, which generates samples only when necessary. This approach seeks to enhance the representation of underrepresented classes while minimizing the influence of majority class bias. By employing adaptive up-sampling, the method reduces dependence on generated samples, thereby achieving a balanced improvement across class distributions. Ultimately, this study seeks to bridge the gap between the promising potential of GAN-based data augmentation and the practical challenges posed by class imbalance in land cover classification.

The contributions of this paper are as follows:

- Adaptive generation technique that uses the size of the minority class as a dynamic scaling factor, ensuring that only the requisite number of samples are generated, thereby avoiding overgeneration and synthetic bias.
- addresses synthetic bias by generating samples proportionately to the minority class size, preventing unnecessary dataset inflation and preserving the authenticity of rare class representations, leading to more realistic data distributions and improved model performance.
- Enabling more accurate land cover classification by using a balanced dataset, where both rare and majority classes are given equal weight, thereby improving feature extraction capabilities through a 2D-CNN.

By systematically exploring the use of cGANs to address class imbalance, this research aims to contribute to the development of more robust and accurate land cover classification systems, facilitating better decision-making in environmental and urban management.

The remainder of this paper is structured as follows. Section II reviews the related work on imbalanced dataset classification and the use of GANs in data augmentation. Section III describes the methodology, including the architecture of the cGAN used and the process for generating synthetic samples. Section IV presents the experimental setup, datasets used, and the evaluation metrics. Section V discusses the results and their implications for land cover classification. Finally, Section 6 concludes the paper with a summary of findings and directions for future research.

## II. LITERATURE SURVEY

The classification of LCSs using remote sensing data has been an area of active research for several decades, owing to its significant applications in environmental monitoring, urban planning, and resource management. HSI has become a cornerstone in deep learning models for classification tasks across diverse domains. Recent advancements, including SSTNet [16], GACNet [17], and CATNet [18], highlight innovative approaches that exploit the spatial, spectral, and texture features of HSI. SSTNet integrates 3-D and 2-D convolutions with attention mechanisms to capture both spectral and spatial details effectively. GACNet enhances classification through semi-supervised GANs for data augmentation and cross-aware attention for feature refinement. CATNet introduces cascaded attention and transformer-based modules to handle the complex feature spaces of HSI. These models address key challenges such as data scarcity, redundancy, and noise, pushing the boundaries of HSI-based classification in terms of accuracy and robustness.

Advanced models such as HybridSN, SSRN, and SpectralFormer have significantly improved the effectiveness of HSI by addressing the complexities associated with hyperspectral data. HybridSN utilizes both 2-D and 3-D convolutional layers, effectively capturing spectral and spatial features from HSI, which is essential for handling complex land cover data [19]. SSRN adopts a residual learning framework that refines feature extraction by emphasizing the interdependence between the spectral and spatial dimensions of HSI, leading to more accurate classifications [20]. SpectralFormer, incorporating transformer-based architectures, utilizes self-attention mechanisms to capture

long-range dependencies within hyperspectral data, further improving classification performance [21].

Despite the advancements in classification algorithms, the issue of class imbalance remains a persistent challenge, impacting the accuracy and robustness of classification models. Class imbalance occurs when certain classes are underrepresented in the training dataset, leading to biased models that perform poorly on minority classes. Traditional approaches to mitigate this problem, such as oversampling, undersampling, and cost-sensitive learning, have shown limited success, often leading to issues like overfitting or loss of information.

In recent years, GANs have emerged as a promising solution to address the limitations of conventional methods. By generating synthetic data, GANs can effectively balance the class distribution without the drawbacks associated with traditional techniques. cGANs, an extension of GANs, offer the capability to generate samples conditioned on specific class labels, making them particularly suited for addressing class imbalance in classification tasks. This section reviews the current state of research on LCSs classification, the impact of class imbalance, traditional methods for handling imbalance, and the application of cGANs, in generating synthetic data to enhance classification performance.

### A. GAN BASED SYNTHESIZING OF HSI DATA

GANs have emerged as a prominent tool for generating synthetic samples in the domain of HSI analysis, exhibiting considerable potential in addressing the challenge of limited sample availability. Extensive research in this area has yielded promising outcomes, with GANs showcasing efficacy in augmenting datasets for improved performance of HSI-based applications. A comprehensive review of relevant literature has been conducted, wherein several research studies have been examined to elucidate the utilization of GANs for HSI sample generation. The literature review section provides detailed analyses of these studies, contributing to a deeper understanding of the current landscape and advancements in the field. Jamali et al. [6] used a 3-D hybrid GAN that Combine GAN with Hybrid CNN that uses both 2D and 3D features. Zhang et al. [15] used a progressive growing GAN combined with a Wasserstein generative adversarial network. Wang et al. [14] proposed DcCapsGAN which is a Dual-channel capsule generative adversarial network for HSI classification. They constructed a dual channels GAN that uses 1-D and 2-D GAN. Shi et al. [11] proposed IGAN, which is a simple GAN for very high-resolution images. Liang et al. [7] proposed SSAT-GAN that synthesizes data using GAN and then enhances their description by using the Spectral Spatial Attention Module. Song et al. [12], proposed TBGAN that utilizes two branch GAN, where the generator extracts spatial and spectral features respectively. Further, they used skip connections to avoid vanishing gradient. Hao et al. [5] proposed TRUG model which is a Transformer based method to deal with spectral features. They used GAN with a transformer module for HSI classification.

### B. GAN FOR DEALING WITH CLASS IMBALANCE SAMPLES

The collective body of research in this domain primarily centers on the generation of HSI, with a particular emphasis on spectral, spatial, or spectral-spatial synthesis methodologies. However, within this spectrum of studies, a subset of the literature delves specifically into addressing the challenge of class imbalance, which bears direct relevance to the objectives of our investigation. This subset of scholarly works explores strategies and techniques aimed at mitigating class imbalance issues within HSI datasets, thereby aligning closely with the thematic focus of our research endeavor. Wang et al. [13] propose a method for improving the classification of HSI using a special type of GAN called ADGAN (Adaptive DropBlock-enhanced GAN). Reference [9] propose a DCGAN framework for Hyperspectral Image Classification that utilizes convolutional autoencoders and distance constraints to improve classification accuracy, particularly for datasets with limited labeled samples and class imbalance. He et al. [4] introduced HyperViTGAN, a semi-supervised generative adversarial network incorporating transformers specifically designed to classify hyperspectral images by generating training samples for under-represented classes during the training phase and utilizing 3D image patches. Feng et al. [2] propose a self-supervised approach called SDC-GAN that combines GANs with clustering to improve the classification of hyperspectral images that have imbalanced class distributions and limited labeled data.

### C. CGAN BASED SYNTHESIZING OF HSI DATA

While various techniques have been explored to address imbalanced class problems, cGAN-based approaches have emerged as a particularly promising avenue of research. Roy et al. [10] 3D-HyperGAMO, which utilizes a GAN during training to generate synthetic 3D spectral-spatial patches for minority classes in hyperspectral image classification, aims to improve classification accuracy by balancing the dataset. Dam et al. [1] proposed MGSGAN, which is a mixture of Spectral Generative Adversarial Networks, addresses imbalanced class distributions in hyperspectral image classification by employing a GAN architecture that enforces generated samples to adhere to the true class distributions.

The preceding review provides a concise overview of relevant literature addressing class imbalance problems. A brief summary of these studies is presented in Table 1 for further reference.

Despite significant research on applying GANs to HSI classification, notable gaps still exist. Many previous studies have focused on generating class-specific data using cGANs but have overlooked the critical issue of synthetic bias, particularly for rare classes. These models typically rely on the largest class as a scaling factor, leading to the overgeneration of samples for rare classes. This overgeneration introduces

**TABLE 1.** This table provides a detailed synthesis of critical aspects derived from the reviewed literature. It encompasses key references, datasets employed, methodologies and techniques applied, primary findings, identified research gaps, and the nature of the data used. The table serves as a valuable resource for understanding the methodologies, outcomes, and contributions of existing studies in the field, while also highlighting areas that require further exploration. This overview not only contextualizes current research but also identifies emerging opportunities for advancing the domain.

| No | Reference | Methodology | Dataset | Metric | Findings | Gap |
|---|---|---|---|---|---|---|
| 1 | [6] *et al.* | Combine 3D-GAN with CNN to synthesize samples for small classes only. | Rapid Eye Image of Avalon pilot site | Kappa, OA, AA | "Increased samples improved scores." | Model not tested on diverse datasets, ignores data imbalance problem. |
| 2 | [15] *et al.* | progressive growing GAN and Wasserstein GAN approach | Indian Pines and University of Pavia | Kappa, OA, AA | Improved performance and training process. | Ignores data imbalance in a highly complex model. |
| 3 | [14] *et al.* | Combining 1D and 2D CapsNet with GAN. | University of Pavia, Salinas, Kennedy Space Center | Kappa, OA, AA | Accuracy improved, but complexity increased with the Capsule structure. | Complexity |
| 4 | [11] *et al.* | Simple GAN for VHR Very High Resolution Images | Aerial image data, Vaihingen image data, VHR Images | Kappa, OA, AA | Increased accuracy but low performance in some categories | Suboptimal sampling, disregarded class imbalance. |
| 5 | [7] *et al.* | Generate HSI using GAN, then enhance with Spectral Spatial Attention Module and four proposed models. | Indian Pines, Pavia University | Kappa, OA, AA | Improved performance using attention-based spectral-spatial feature learning. | System overhead addressed; class imbalance ignored. . |
| 6 | [12] *et al.* | The generator in the Two-Branch GAN uses skip connections to prevent vanishing gradients while extracting features. | Indian Pines, Salinas, Pavia University | Kappa, OA, AA | Enhanced robustness with reduced sensitivity to training data. | Added dual-branch GAN, ignoring class imbalance. |
| 7 | [5] *et al.* | Transformer based method to deal with spectral features | Indian Pines, Pavia University | Kappa, OA, AA | Transformer based GAN, Improves feature learning | Training mode breakdown, Ignored class Specific data generation. |
| 8 | [5] *et al.* | Address data imbalance and mode collapse using adjusted discriminator and adaptive DropBlock. | Indian Pines, Salinas | Kappa, OA, AA | Enhanced classification through improved feature learning. | Ignored Imbalanced Problem, No Class-Specific Data Generation |
| 9 | [9] | Used Conv AE and Distance constraint on generator and discriminator | Indian Pines, Pavia University, and Botswana | Kappa, OA, AA | Good performance with minimal training data, without explicitly reducing interclass similarity. | Neglected imbalanced problems and class-specific data generation. 3.5 |
| 10 | [4] *et al.* | GAN with transformer and skip connections | Houston, Indian Pines, and Xuzhou, | Kappa, OA, AA | Improved classification performance | Imbalanced Training Data, Class-Specific Data Generation |
| 11 | [2] *et al.* | Divide and Conquer, Clustering based approach | Aihinge, 2018 IEEE GRSS DFC Dataset | Kappa, OA, AA | Reduces mode collapse, enhances sample quality, and improves accuracy. | Imbalanced Training Data, Class-Specific Data Generation |
| 12 | [10] *et al.* | Generation of samples for small class at the time of training. | Indian Pines, Kennedy Space Center, University of Pavia, and Botswana | Kappa, OA, AA | More effective with small training sets with high imbalance | High complexity due to 3D convolution, Imbalanced Training Data |
| 13 | [1] *et al.* | GAN based generation of the samples forcing them to be in the actual dimensions. | Indian Pines, Pavia University | Kappa, OA, AA | Improved Classification Performance | Imbalanced Training Data, Not used spatial Features |

synthetic bias, distorting the representation of rare classes and negatively impacting classification accuracy.

To address this challenge, our approach introduces an adaptive generation technique that dynamically uses the size of the minority class as a scaling factor. This allows for the generation of only the necessary samples, ensuring that rare classes are adequately represented without excessive synthetic bias. By emphasizing rare classes during the synthesis phase, we guarantee that the generated samples more accurately reflect the true data distribution, resulting in improved classification performance.

## III. METHODOLOGY

This study proposes a methodology to mitigate the challenges posed by class imbalance in HSI classification through the application of advanced machine learning techniques, specifically focusing on GAN. The research addresses the fundamental issue of uneven class distribution within HSI datasets, which often leads to biased model performance favoring majority classes at the expense of minority ones. To alleviate this imbalance, our approach focuses on the utilization of cGANs to generate synthetic data samples tailored to underrepresented classes. By augmenting the dataset with these synthesized samples, we aim to enhance the classifier's ability to accurately recognize and classify all land cover types, including those that are rare or less frequently observed. The methodology integrates data preprocessing, cGAN training, and classifier development within a coherent framework designed to improve the robustness and reliability of HSI classification systems in real-world applications.

Our proposed model SB-GAN, draws inspiration from the 3D-HyperGAMO architecture presented by Roy et al. [10], yet it features a reduced patch size and a simplified architectural framework. Unlike the original HyperGAMO, which employs a 3D-Convolutional model marked by its complexity, our model utilizes a 2D-Convolutional architecture. Specifically, we implement a smaller patch size of 9 compared to HyperGAMO's patch size of 11, effectively reducing the input dimensionality and consequently, the scale of the learning parameters involved.

To tackle the challenges associated with limited sample sizes and class imbalance in HSI classification, our approach begins with the pre-processing steps, following by training cGAN model that is deployed to fabricate synthetic, class-specific samples. After the generation of these samples, they are processed through a 2D Convolutional Neural Network (2D-CNN) classifier. This methodology not only facilitates the classification of balanced and synthesized HSI data but also enhances classification accuracy by optimally leveraging both authentic and artificially generated datasets in a harmonized manner. This strategic integration of real and synthetic data ensures a robust performance boost in the classification process. The abstract view of the model is shown in Figure 1.

The overall structure of the model is explained in the below sections:

### A. 3D PATCH EXTRACTION

An HSI can be denoted as a three-dimensional tensor, where $W, H$, and $B$ represent the width, height, and number of spectral bands, respectively. It is defined as $HSI_{\text{orig}} \in \mathbb{R}^{W\times H\times B}$. The classification task for HSI typically involves assigning each pixel to one of $L_c$ land-cover classes, represented by $Y = (y_1, y_2, \ldots, y_{L_c})$. Each pixel $HSI_{i,j} \in HSI_{\text{orig}}$, where $i = 1, \ldots, W$ and $j = 1, \ldots, H$, is characterized by a spectral vector $x_{i,j} = [d_{i,j,1}, \ldots, x_{i,j,d}] \in \mathbb{R}^d$, representing the spectral profile of the observed surface materials.

To extract 3D patches from the HSI data, patches of dimensions $p \times p \times d$ are generated, where $p \times p$ denotes the spatial dimensions and $d$ is the number of spectral channels. A sliding window approach with a defined stride of $s$ is employed across each channel over the spatial domain to produce the 3D patches. These patches serve as inputs to a downsampling unit, to make the dataset balanced.

For the purpose of majority downsampling within a deep learning framework, where effective utilization of both spectral and spatial information is desired, all extracted patches are stacked into $D = \{x_1, x_2, \ldots, x_i, \ldots, x_n\}$, where $d_i \in \mathbb{R}^{p\times p\times d}$ and $n$ denotes the total number of patches. Empirically, a spatial window size of $p \times p \times d$ is defined, set to $9 \times 9 \times$ Dimensions across the three datasets considered in our experiments.

### B. CGAN ARCHITECTURE

cGAN is an extension of the basic GAN framework, designed to generate data conditioned on some auxiliary information such as class labels, text descriptions, or another modality of data. This condition enhances the capability of the network to control aspects of the generated data, making the model more versatile and targeted in its application.

The cGAN model consists of two main components: the generator (G) and the discriminator (D). The generator's role is to produce synthetic data that closely resembles the real data, conditioned upon some external information. For example, in an image patch generation task, the generator might create patches of a specific class using a predetermined label provided as input.

The discriminator, conversely, is trained to distinguish between the actual data and the synthetic data produced by the generator, also taking into account the same conditional information. The discriminator evaluates the authenticity of the combined input (real or synthetic data and condition) and attempts to ascertain whether the data is genuine or fabricated.

Training a cGAN involves a minimax game where the generator tries to maximize the probability of the discriminator making a mistake (thus becoming better at generating realistic data), while the discriminator aims to minimize this probability by getting better at distinguishing real from fake. This adversarial process continues until a point of equilibrium is reached where the generator produces data indistinguishable from real data according to the discriminator, effectively learning the data distribution conditioned on the auxiliary information. The cGAN model takes input as the class label and concatenates that label with the input patch. The input is an image patch when using a discriminator and it is $z$ noise when using a generator. The concatenation phase is followed by 2D convolutional layer in the discriminator and 2D transposed convolutional layers in the generator module.

The objective function of a cGAN plays a pivotal role in dictating how the generator and discriminator learn to synthesize and differentiate, respectively, conditioned on additional information such as class labels or data from other modalities. A cGAN network modifies the traditional GAN framework to condition both the generator $G$ and the discriminator $D$ on some external data $y$, enhancing the ability to direct the data generation process. The objective function of a cGAN can be represented as:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x|y)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z|y)))]$$

$\mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x|y)]$ represents the expected log-probability that the discriminator $D$ correctly identifies real data $x$ as actual, given the condition $y$. Here, $p_{\text{data}}$ is the probability distribution of the real data. The goal of the discriminator is to maximize this probability (hence the log in the function), indicating that it can recognize real instances with high confidence. $z$ in equation $\mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z|y)))]$, is a noise vector sampled from some probability distribution $p_z$, typically a Gaussian distribution.

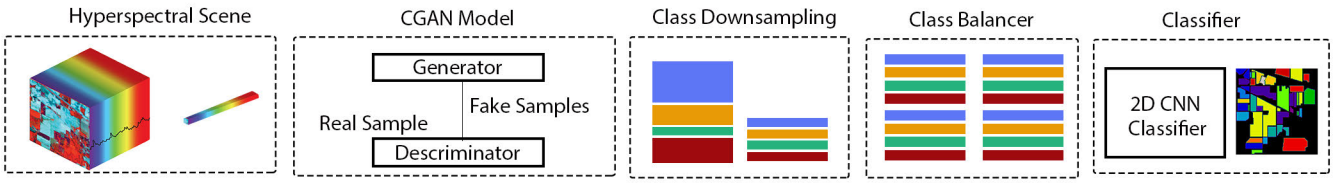



**FIGURE 1.** Abstract view of the proposed model, consisting of four distinct modules: (1) Data Preparation, (2) Downsampling, (3) cGAN Training, (4) Data Augmentation, and (5) Classification. Each module plays a crucial role in the pipeline, where the HSI samples are first converted to small patches, then the patches are downsampled to mitigate majority class bias-ness, followed by the training phase of cGAN model and then the generation of synthetic data to augment the dataset, and finally classified to derive meaningful insights.

$G(z|y)$ denotes the output of the generator, conditioned on $y$. This output is a synthetic instance meant to mimic the distribution of real data. The discriminator $D$ tries to minimize this value by assigning a low probability to the fake data produced by the generator, which is reflected by $\log(1 - D(G(z|y)))$. The discriminator $D$ seeks to maximize the function $V(D, G)$ to correctly classify both real and generated data, while the generator $G$ aims to minimize it by trying to 'fool' the discriminator into mistaking its outputs (fake data) for real data. This adversarial setup encourages both networks to improve continuously, enhancing the quality of the generated samples and the accuracy of the discriminator.

This objective function essentially creates a dynamic game between the two networks, with the generator improving its capability to produce increasingly realistic data (as judged conditional on $y$), and the discriminator becoming better at distinguishing real data from generated data. In our proposed model, cGAN is trained over a downsampled dataset that has the better discriminative ability and thus is not based on the majority class. After training the cGAN, the generator module is used to generate synthetic data in the same amount used for training, thus doubling the size of the balanced data used for training. It is possible to generate additional data; however, experimental results indicate that doubling the dataset size yields satisfactory outcomes. Consequently, there is no need to impose additional computational overhead on the system.

### C. BALANCING, BY ADAPTIVE MINIMUM-DRIVEN CLASS BALANCER

Following the training phase of the cGAN model, a new sub-dataset is created for the 2D-CNN classifier. In this process, the rare classes from the original dataset are oversampled by generating additional samples using the trained generator. The size of the balanced dataset is effectively doubled through this synthetic augmentation. For classes that already have a sufficient number of samples, the existing samples are retained as they are. The remaining required samples are generated using the cGAN model. Consequently, all classes in the dataset attain double the number of samples compared to the balanced dataset utilized for cGAN training.

This module is designed to adaptively balance the class distribution by focusing on the minority class size and ensuring that only the necessary number of samples are generated, avoiding both synthetic bias and dataset inflation. The formulation of this problem is as follows:

#### 1) IDENTIFY THE MINORITY CLASS SIZE

Let $D$ denote the original hyperspectral image (HSI) dataset, consisting of $k$ image patches:

$$D = \{(x_i, y_i)\}_{i=1}^{k}$$

where $x_i$ represents the 3D HSI patch features and $y_i$ is the corresponding class label. The number of samples in each class $k$ is denoted as $n_k$ for $k = 1, 2, \ldots, K$. The minimum class size is defined as:

$$n_{\min} = \min_k n_k$$

This defines the smallest class size, which will be used as the scaling factor for balancing other classes.

#### 2) COMPUTE THE REQUISIT NUMBER OF SAMPLES FOR EACH CLASS

The requisit number of samples $n_{\text{requisit}}$ for each class is calculated by scaling the minority class size $n_{\min}$ with a scaling factor $s$:

$$n_{\text{requisit}} = n_{\min} \times s, \quad \forall k \in \{1, 2, \ldots, K\}$$

This determines how many samples each class should have to achieve balance.

#### 3) COMPUTE THE IMBALANCE FOR EACH CLASS

For each class $k$, the imbalance is determined by the difference between the requisit number of samples and the actual class size:

$$\Delta_k = n_{\text{requisit}} - n_k$$

- If $\Delta_k > 0$, the class $k$ is underrepresented, and additional samples need to be generated.
- If $\Delta_k \leq 0$ , the class $k$ is balanced or overrepresented, and no further sampling is required.

#### 4) RANDOM SAMPLING FOR MAJORITY CLASSES

For classes where $\Delta_k \leq 0$ (balanced or overrepresented classes), we perform random sampling to retain a diversity of samples:

$$D_k^{\text{down}} \subset D_k, \quad \text{such that} \quad |D_k^{\text{down}}| = n_{\text{requisit}}, \quad \text{if } \Delta_k \leq 0$$

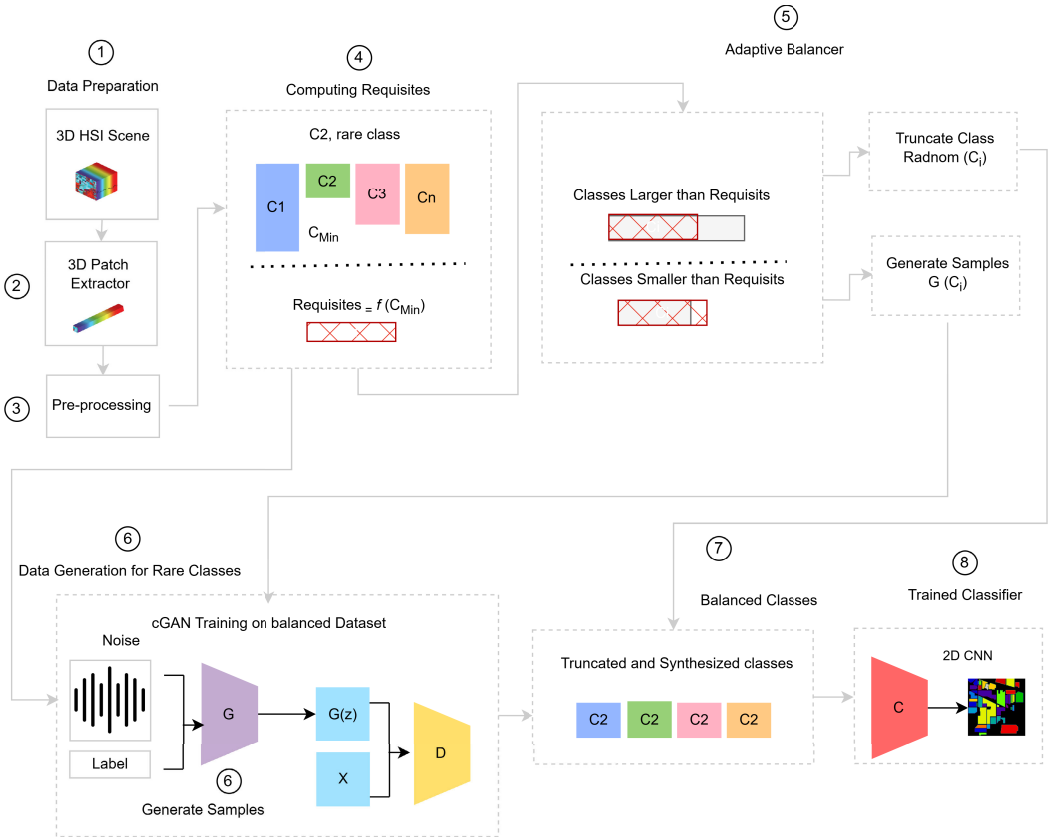



**FIGURE 2.** A detailed view of the proposed SB-GAN model, which consists of seven sequential modules: (1) Data Input (3D HSI Cube), (2) 3D Patch Extraction, (3) Preprocessing (normalization, background removal, data normalization, and PCA application), (4) Computing Requisites, (5) Adaptive Balancer, (6) cGAN Training, (7) Data Augmentation, and (8) Classification using a 2D CNN, followed by the generation of a Classification Map. In the Data Input phase, the 3D HSI cube is introduced into the pipeline. The 3D Patch Extraction module divides the data into smaller, manageable patches. During the Preprocessing phase, the data undergoes normalization, background removal, and PCA to reduce dimensionality. The Requisites computing phase involves the number of required samples for each class irespective of the samples present in a class, followed by an adaptive balancer that decided how many samples to generate for each class. In the cGAN Training phase, both the label and patches are fed into the conditional GAN model to train it to generate realistic data. The subsequent Data Augmentation phase involves combining the original data samples with the synthetic data generated by the cGAN for rare classes. This results in an enhanced, balanced dataset that mitigates the imbalance in rare classes, improving accuracy. Finally, in the Classification phase, a 2D CNN is used to classify the augmented dataset, producing a classification map that provides valuable insights into the spatial distribution of the classes.

where $D_k$ is the subset of $D$ corresponding to class $k$, and $D_k^{\text{down}}$ represents the randomly selected subset.

#### 5) GENERATION OF SAMPLES FOR UNDERREPRESENTED CLASSES

For classes where $\Delta_k > 0$ (underrepresented classes), we need to generate additional samples. The remaining number of samples to be generated is:

$$\text{Remaining Samples}_k = \Delta_k$$

These remaining samples are generated using a generative model $G$ (e.g., a cGAN), which produces synthetic samples:

$$D_k^{\text{gen}} = G(\text{Remaining Samples}_k), \quad \text{if } \Delta_k > 0$$

where $D_k^{\text{gen}}$ represents the synthetic samples generated for class $k$.

#### 6) CONSTRUCT THE FINAL BALANCED DATASET

The final balanced dataset $D^{\text{balanced}}$ is obtained by combining the downsampled and generated samples from each class:

$$D^{\text{balanced}} = \bigcup_{k=1}^{K} D_k^{\text{down}} \cup D_k^{\text{gen}}$$

This balanced dataset is then used for training the GAN model, ensuring that class imbalances do not adversely affect the model's performance.

### D. 2D CLASSIFIER

After synthesizing additional data using a conditional Generative Adversarial Network (cGAN), a 2D Convolutional Neural Network (2D-CNN) model is incorporated to perform the actual classification. This process begins by augmenting the dataset, specifically oversampling the minor classes to address class imbalance. The newly synthesized samples are

combined with the original samples to create an enlarged, balanced dataset that has doubled the size of the original. This comprehensive dataset, now containing balanced samples in each class, is prepared for training the classifier.

The 2D-CNN classifier is trained on this concatenated dataset, which includes both the original balanced set and the synthesized balanced set. The classifier takes input patches from the enlarged dataset, which are preprocessed appropriately for the model. The architecture of the 2D-CNN consists of two modules, each comprising a Conv2D followed by a Leaky Rectified Linear Unit (LeakyReLU) activation function. This structure allows the model to effectively learn and extract features from the input data.

At the final stage of the model, the feature maps generated by the convolutional layers are flattened into a single vector. This flattened feature map is then used for classification. The model optimization is performed using the Adam optimizer, which adjusts the learning rate adaptively and efficiently converges to a minimum. The categorical cross-entropy loss function is employed to measure the performance of the model, as it is well-suited for multi-class classification problems.

By training the 2D-CNN on this robust and balanced dataset, the classifier is better equipped to generalize from the data and achieve more accurate classification results. The use of cGAN for data synthesis and the carefully designed CNN architecture contribute to the improved performance and reliability of the model. The architecture of the classifier is given in Figure 2 and the above steps are given in Algorithm 1.

The layer detail of D-Path-AE model is given in Table 2.

### E. IMPLEMENTATION, DETAILS

As a preprocessing step, we have applied Principal Component Analysis (PCA) to reduce the dimensions of HSI cube and to extract useful features from the HSI cube. HSI has continuous spectral bands and therefore contains redundant information. We train a cGAN model on real and Fake samples generated using a generator module using a noise $Z$. We explicitly trained the cGAN to learn to generate fake patches for HSI data. To add non-linearity, we applied the LeakyReLU function to learn complex non-linear features of the spectra. The leaky ReLU (Rectified Linear Unit) function is a variant of the standard ReLU function, designed to address the "dying ReLU" problem by allowing a small, non-zero gradient for negative input values. The leaky ReLU function can be mathematically defined as:

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases}$$

where $\alpha$ is a small positive constant, typically set to a small value such as 0.01. This ensures that even for negative input values, there is a non-zero gradient, preventing the neuron from being completely inactive. Logistic Activation also known as sigmoid activation is straightforward and the base of many GANs and CNN models. The sigmoid is a differentiable monotonically increasing function that takes any real-valued number and maps it to $[0, 1]$. For large

**Algorithm 1** SB-GAN Algorithm

**Input:** HSI scene $\mathbf{HSI_{Orig}} \in \mathbb{R}^{h \times w \times d}$
Learn Features thorugh PCA: $\mathbf{D}_{\text{PCA}} = \text{PCA}(\mathbf{HSI_{Orig}}, n_dim)$
Extract 3D patches from $\mathbf{D}_{\text{PCA}}$:

$$\mathbf{D} = \{\mathbf{D}_i \in \mathbb{R}^{p \times p \times d} \mid i = 1, 2, \ldots, n\}$$

$\mathbf{X}_{\text{train}}, \mathbf{X}_{\text{test}} = \text{train_test_split}(\mathbf{X}, \text{test_size} = t)$
Initialize cGAN with generator $G$ and discriminator $D$
Define loss functions for $D$ and $G$:

$$\mathcal{L}_D = -\mathbb{E}_{\mathbf{x} \sim \mathbf{P}_{\text{real}}}[\log D(\mathbf{x})] - \mathbb{E}_{\mathbf{z} \sim p_z}[\log(1 - D(G(\mathbf{z})))]$$
$$\mathcal{L}_G = -\mathbb{E}_{\mathbf{z} \sim p_z}[\log D(G(\mathbf{z}))]$$

**for** epoch in range(epochs) **do**
  **for** each batch $\mathbf{B}_{\text{real}}$ in $\mathbf{X}_{\text{train}}$ **do**
    Generate fake patches: $\mathbf{B}_{\text{fake}} = G(\mathbf{z})$: $\mathbf{z} \sim X_z$, Sample noise
    Update $D$: $\mathcal{L}_D \leftarrow -\mathbb{E}[\log D(\mathbf{B}_{\text{real}})] - \mathbb{E}[\log(1 - D(\mathbf{B}_{\text{fake}}))]$
    $D \leftarrow \text{optimizer}_D(\mathcal{L}_D)$
    Generate fake patches: $\mathbf{B}_{\text{fake}} = G(\mathbf{z})$: $\mathbf{z} \sim p_z$, Sample noise
    Update $G$: $\mathcal{L}_G \leftarrow -\mathbb{E}[\log D(\mathbf{B}_{\text{fake}})]$
    $G \leftarrow \text{optimizer}_G(\mathcal{L}_G)$
  **end for**
**end for**
Identify smallest class $C_{\min}$ with $n_{\min}$ samples
$s = 2$ Scaling Factor,
$Requisits = n_{\min} xs$
**for** each class $C_i$ **do**
  **if** $[(\text{Number of samples in } C_i) - Requisits] \geq 0$ **then**
    Select random $Requisits$ samples from $C_i$
  **else**
    Select all real samples from $C_i$
    Generate $Requisists - C_i$ samples using $G$
  **end if**
**end for**
Concatenate all real and generated samples to form $\mathbf{D}_{\text{balanced}}$:

$$\mathbf{D}_{\text{balanced}} = \bigcup_i \mathbf{S}_i$$

**for** epoch in range(epochs) **do**
  **for** each batch $\mathbf{B}_{\text{balanced}}$ in $\mathbf{P}_{\text{balanced}}$ **do**
    Compute loss: $\mathcal{L}_{\text{CNN}} \leftarrow \text{loss_function}(\text{CNN}(\mathbf{B}_{\text{balanced}}))$
    Update CNN: $\text{CNN} \leftarrow \text{optimizer}(\mathcal{L}_{\text{CNN}})$
  **end for**
**end for**

**TABLE 2.** This table presents the architectural details of the cGAN Discriminator, cGAN Generator, and 2D CNN Classifier. For each model component, the table specifies the layer type, its preceding layer, and the corresponding label. The cGAN Discriminator incorporates dense and reshape layers to combine the label with the patch size, followed by a concatenation of label and patch features. It then applies two Conv2D layers, followed by flattening and additional dense layers. The cGAN Generator starts with input noise, passing through dense and reshape layers to align the input with the patch size, which is then concatenated with the label and patch features in a similar manner to the discriminator. The generator proceeds with three stacked Conv2DTranspose layers for upsampling. The 2D CNN Classifier includes two Conv2D layers, followed by a flattening layer and a dense layer, with a softmax output for classification. LeakyReLU activation is utilized in all Conv2D and Conv2DTranspose layers of the model. The table illustrates the sequential connectivity and flow of information across all layers of the models.

| Layer Type | Connected to | Activation | Label |
|---|---|---|---|
| **Discriminator** | | | |
| Input(1) | | | Input_1 |
| Dense(w,w) | Input_1 | Linear | D_1 |
| Reshape(w,w, 1) | D_1 | | |
| Input(w,w,d) | | | Input_2 |
| Concat | D_1, Input_2 | | merge |
| Conv2D(32) | merge | LeakyReLU | Conv_1 |
| Conv2D(64) | Conv_1 | LeakyReLU | Conv_2 |
| Flatten | Conv_2 | | F_1 |
| Dense(1) | F_1 | Sigmoid | |
| **Generator** | | | |
| Input(1) | | | Input_1 |
| Dense(w,w) | Input_1 | Linear | D_1 |
| Reshape(w,w, 1) | D_1 | | |
| Input(Latent_Dim) | | | Input_2 |
| Dense(200,w,w) | Input_2 | LeakyReLU | D_2 |
| Reshape(200, w,w) | D_2 | | R_1 |
| Concat | R_1, Input_1 | | merge |
| Conv2DTranspose(Dim*3) | merge | LeakyReLU | conv_T_1 |
| Conv2DTranspose(Dim*2) | conv_T_1 | LeakyReLU | conv_T_2 |
| Conv2DTranspose(Dim*1) | conv_T_2 | LeakyReLU | conv_T_3 |
| **Classifier** | | | |
| Input(w,w,d) | | | Input_1 |
| Conv_2d(32) | Input_1 | LeakyReLU | Conv_1 |
| Conv_2d(64) | Conv_1 | LeakyReLU | Conv_2 |
| Flatten | Conv_2 | | F_1 |
| Dense(N_Classes) | F_1 | Softmax | |

negative numbers, it asymptotes to 0 while for large positive numbers, it asymptotes to 1. It is defined as

$$sigmoid(\mathbf{z}) = \frac{1}{1 + \exp(-\mathbf{z})}. \tag{1}$$

## IV. RESULTS AND EXPERIMENTS

This section provides a comprehensive examination of the datasets and experimental procedures used to evaluate the performance of various machine learning (ML) models. The models under study include traditional approaches such as Logistic Regression (LR), Decision Tree (DT), and Support Vector Machine (SVM), as well as advanced deep learning models like Multi-Layer Perceptron (MLP), Convolutional Neural Networks (CNN), AD-GAN, HybridSN, SSRN, SF and SB-GAN. The analysis encompasses a thorough evaluation of performance metrics, such as accuracy, precision, recall, and F1-score, as well as runtime efficiency, to assess the computational demands of each model. Additionally, we analyze the classification maps generated by these models to visualize and interpret their performance in different scenarios. All experiments were conducted on a machine equipped with an Intel(R) Core(TM) i5-8350U CPU running at 1.70GHz to 1.90GHz, with 16 GB of RAM. This setup ensures that the computational resources are adequate for running both traditional ML and deep learning models, allowing for a consistent comparison of their performance under similar conditions.

### A. DATASET

HSI data is three-dimensional data containing spatial and spectral features and is stored in a 3-D data cube $(h, w, d)$, where h,w represents the spatial coordinates while $d$ represents the spectral details. $d$ is a vector that measures the spectral bands at a specific pixel $h, w$. We have used three publicly available datasets discussed as under:

#### 1) INDIAN PINES DATASET

The Indian Pines dataset is a valuable resource for studying land cover using hyperspectral imagery. It encompasses a region in North-western Indiana, captured across 145 × 145 pixels and 220 spectral bands, providing information on reflectance at various wavelengths (0.4 – 2.5 micrometers). Each pixel is labeled with one of sixteen land cover types, making it suitable for image segmentation tasks. IP dataset is used for research in hyperspectral image classification, land cover mapping, and the development of machine learning algorithms for analyzing this type of remote sensing data. Table 3 shows the details of the samples and classes in this dataset. The ground truth and RGB scene of the dataset are shown in Figure 3

#### 2) PAVIA UNIVERSITY DATASET

The Pavia University dataset is a cornerstone for research in HSI classification. Captured by the ROSIS-3 sensor over Pavia, Italy, it offers a detailed picture of the region. The data consists of a 610 × 340 pixel image, containing information from 115 spectral bands. These bands capture the reflectance of various materials within the scene across a wide range of wavelengths, typically from 430 to 860 nanometers (visible and near-infrared). This detailed spectral information allows researchers to differentiate between land cover types with high accuracy. Additionally, the dataset provides ground truth labels for each pixel, classifying it into one of 9 categories like asphalt, meadows, trees, or bare soil. This labeled data makes it perfect for training and evaluating classification models. Due to its public availability, the Pavia University dataset remains a valuable resource for researchers in remote sensing and image analysis. Table 3 shows the details of the samples and classes in this dataset. The ground truth and the RGB scene of the dataset are shown in Figure 4.

#### 3) SALINAS DATASET

The Salinas scene dataset was captured by an airborne hyperspectral sensor over the Salinas Valley in California. This dataset features a spatial resolution of 512 × 217 pixels and includes 204 spectral bands, providing a comprehensive view of the spectral properties of the land cover. The spatial

**TABLE 3.** This table provides the dataset details for the Pavia University (PU), Salinas (SA), and Indian Pines (IP), datasets. It includes the name of each class, the number of samples in each class, and the corresponding color assigned to each class. The last row of the table shows the number of samples in the minority class. These details offer insights into the class distribution and visualization aspects of the datasets used in this study.

| Number | Pavia University | Color | Salinas | Color | Indian Pines | Color |
|---|---|---|---|---|---|---|
| 1 | Asphalt | | Broccoli-Green-Weeds-1 | | Alfalfa | |
| 2 | Meadows | | Broccoli-Green-weeds-2 | | Corn-NoTill | |
| 3 | Gravel | | Fallow | | Corn-MinTill | |
| 4 | Trees | | Fallow-Rough-plow | | Corn | |
| 5 | Metal Sheets | | Fallow-Smooth | | Soybeans-NoTill | |
| 6 | Bare Soil | | Stubble | | Soybeans-MinTill | |
| 7 | Bitumen | | Celery | | Soybeans | |
| 8 | Bricks | | Grapes-Untrained | | Wheat | |
| 9 | Shadow | | Soil-Vinyard-Develop | | Woods | |
| 10 | | | Corn-Senesced-Green-Weeds | | Pasture | |
| 11 | | | Lettuce-Romaine-4wk | | Alfalfa | |
| 12 | | | Lettuce-Romaine-5wk | | Sugarbeets | |
| 13 | | | Lettuce-Romaine-6wk | | Oats | |
| 14 | | | Lettuce-Romaine-7wk | | Other | |
| 15 | | | Vinyard-Untrained | | SowedBare | |
| 16 | | | Vinyard-Vertical-Trellis | | Stone-SteelTowers | |
| C_Min | | 947 | | 916 | | 20 |

resolution of the images is approximately 3.7 meters per pixel, allowing for detailed observation of the landscape. To enhance the quality of the data, the first 20 spectral bands, which primarily correspond to regions affected by water absorption and are thus less informative for land cover classification, have been excluded from the dataset. This preprocessing step helps in focusing on the more relevant spectral information. The dataset's detailed characteristics, including the number of samples and the classes represented, are summarized in Table 3. This table provides an overview of the dataset's composition, including the specific classes present and their distribution, which is crucial for understanding the scope and application of the dataset in various analytical tasks. The RGB scene and the ground truth of the scene consist of 16 classes that are shown in Figure 5.

### B. EXPERIMENTAL SETUP

For patch creation, a window of size $9 \times 9$ is used because it gives the highest accuracy and in some cases gives approximately the same accuracy as $11 \times 11$ thus reducing system overhead and giving only few learnable parameters as compared to the window of size $11 \times 11$ which takes more time and processing power. For evaluation purposes, Cohen's Kappa ($\kappa$) measure is used that compute the score by comparing actual and predicted classes and measuring the observed agreement. It is best used for imbalance class problems, as the datasets involved possess an imbalance nature, the ($\kappa$) is one of the good choices and is commonly used for HSI classification in literature. It is calculated using the following formula:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{2}$$

where $p_o$ is the relative observed agreement ratio, $p_e$ is the hypothetical probability of chance agreement. The value of $\kappa$ indicates the level of agreement. $\kappa = 1$ shows perfect agreement, $\kappa = 0$, No agreement and $\kappa < 0$, Less agreement than expected by chance. Another accuracy measure that we use in our experiments is Overall Accuracy (OA). The OA accuracy is the standard measure used in the literature that computed the percentage of making correct decisions. The OA is calculated using the formula: OA and AA are commonly used metrics for evaluating classification performance. Overall Accuracy is defined as the ratio of correctly classified samples across all classes to the total number of samples, and it is expressed as:

$$OA = \frac{\sum_{i=1}^{N} TP_i}{\sum_{i=1}^{N} P_i}, \tag{3}$$

where $TP_i$ represents the number of true positives for class $i$, $P_i$ is the total number of samples in class $i$, and $N$ is the total number of classes. In contrast, Average Accuracy calculates the mean accuracy of all classes, giving equal weight to each class, and is defined as:

$$AA = \frac{1}{N} \sum_{i=1}^{N} \frac{TP_i}{P_i}. \tag{4}$$

In scenarios where the dataset is balanced, OA and AA become identical and returns the same value. The reason behind this equivalence is that in balanced dataset, each class has the same number of samples, $P$ for each class $i$. In OA,

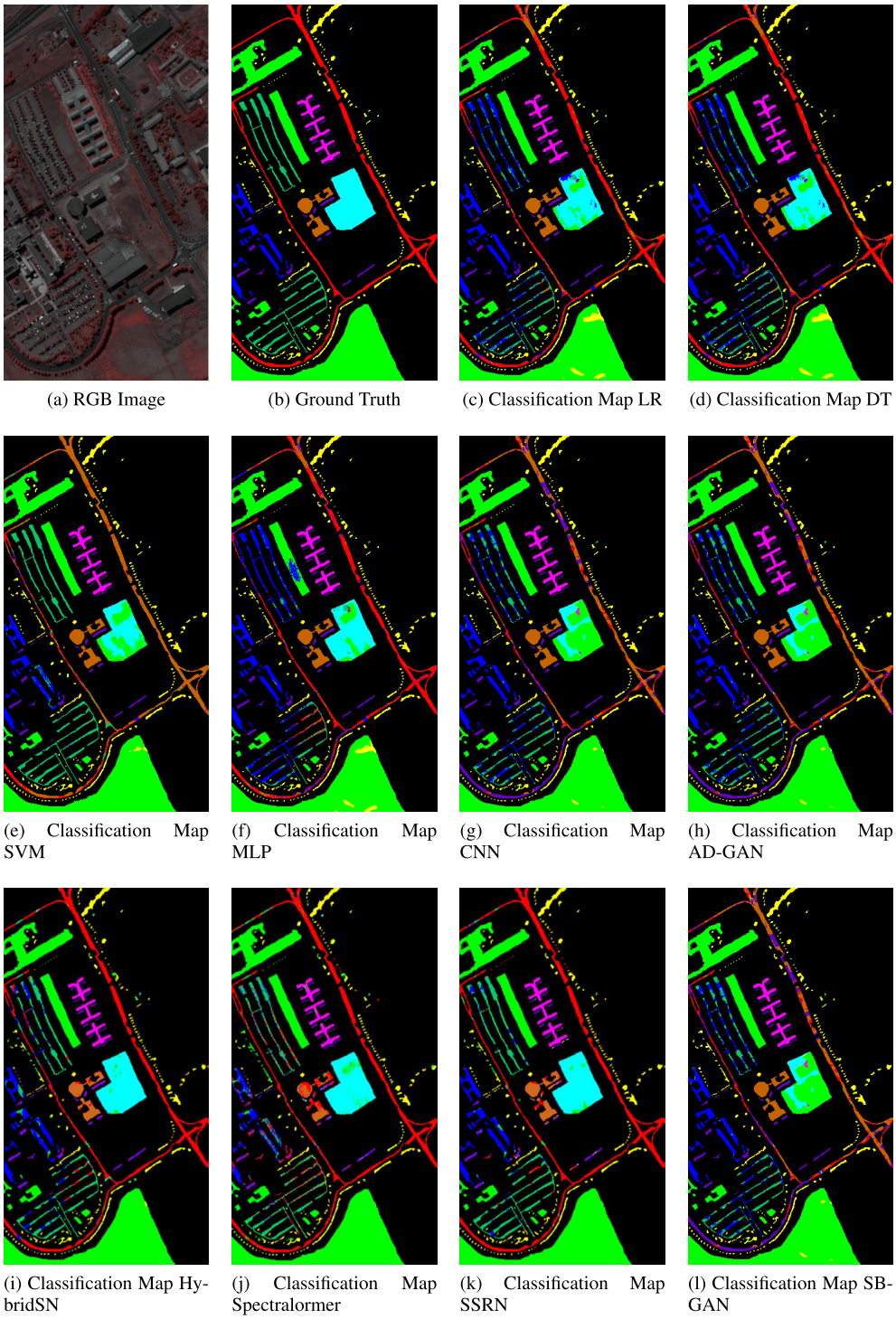
$$\sum_{i=1}^{N} P_i = N \times P, \tag{5}$$

which makes the OA to:

$$OA = \frac{\sum_{i=1}^{N} TP_i}{N \times P}. \tag{6}$$

Similarly, in AA, where each class $P_i$ has same number of samples $P$, the equation becomes:

$$AA = \frac{1}{N} \sum_{i=1}^{N} \frac{TP_i}{P} = \frac{1}{N} \times \frac{1}{P} \sum_{i=1}^{N} TP_i. \tag{7}$$



(a) RGB Image (b) Ground Truth (c) Classification Map LR (d) Classification Map DT

(e) Classification Map SVM (f) Classification Map MLP (g) Classification Map CNN (h) Classification Map AD-GAN

(i) Classification Map HybridSN (j) Classification Map Spectralormer (k) Classification Map SSRN (l) Classification Map SB-GAN

**FIGURE 3. Classification Map of PU Scene: (a) Shows the RGB image of PU using bands (75,70,70). (b) Shows the ground truth of the dataset. (c) Shows the classification map generated through LR classifier. (d) Shows the classification map generated through DT classifier. (e) Shows the classification map generated through SVM classifier. (f) Shows the classification map generated through MLP classifier. (g) Shows the classification map generated through AD-GAN. (h) Shows the classification map generated through HybridSN. (i) Shows the classification map generated through SSRN. (j) Shows the classification map generated through SpectralFormer. (k) Shows the classification map generated through SB-GAN classifier.**

Simplifying further, we obtain:

$$AA = \frac{\sum_{i=1}^{N} TP_i}{N \times P}. \tag{8}$$

This equivalence of OA and AA arises because both metrics normalize the total number of true positives by the same factor, ensuring identical outcomes.

**TABLE 4.** Class-wise Precision, Recall, and F1-Score.

| PU | | | | SA | | | | IP | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Class | Precision | Recall | F1-Score | Class | Precision | Recall | F1-Score | Class | Precision | Recall | F1-Score |
| 0 | 0.99 | 1.00 | 1.00 | 0 | 1.00 | 1.00 | 1.00 | 0 | 0.75 | 1.00 | 0.86 |
| 1 | 0.99 | 1.00 | 1.00 | 1 | 1.00 | 1.00 | 1.00 | 1 | 0.75 | 0.75 | 0.75 |
| 2 | 1.00 | 1.00 | 1.00 | 2 | 1.00 | 1.00 | 1.00 | 2 | 0.75 | 1.00 | 0.86 |
| 3 | 1.00 | 0.99 | 0.99 | 3 | 1.00 | 1.00 | 1.00 | 3 | 1.00 | 1.00 | 1.00 |
| 4 | 1.00 | 1.00 | 1.00 | 4 | 1.00 | 0.99 | 1.00 | 4 | 1.00 | 1.00 | 1.00 |
| 5 | 1.00 | 1.00 | 1.00 | 5 | 1.00 | 1.00 | 1.00 | 5 | 1.00 | 1.00 | 1.00 |
| 6 | 1.00 | 1.00 | 1.00 | 6 | 1.00 | 1.00 | 1.00 | 6 | 1.00 | 0.8 | 0.89 |
| 7 | 0.99 | 0.99 | 0.99 | 7 | 1.00 | 1.00 | 1.00 | 7 | 1.00 | 1.00 | 1.00 |
| 8 | 1.00 | 1.00 | 1.00 | 8 | 1.00 | 1.00 | 1.00 | 8 | 1.00 | 1.00 | 1.00 |
| 9 | | | | 9 | 1.00 | 1.00 | 1.00 | 9 | 0.75 | 0.75 | 0.75 |
| 10 | | | | 10 | 1.00 | 1.00 | 1.00 | 10 | 0.75 | 0.5 | 0.6 |
| 11 | | | | 11 | 1.00 | 1.00 | 1.00 | 11 | 0.25 | 0.5 | 0.33 |
| 12 | | | | 12 | 1.00 | 1.00 | 1.00 | 12 | 1.00 | 1.00 | 1.00 |
| 13 | | | | 13 | 1.00 | 1.00 | 1.00 | 13 | 0.75 | 1.00 | 0.86 |
| 14 | | | | 14 | 0.99 | 1.00 | 1.00 | 14 | 1.00 | 0.67 | 0.8 |
| 15 | | | | 15 | 1.00 | 1.00 | 1.00 | 15 | 1.00 | 1.00 | 1.00 |
| **micro avg** | **1.00** | **1.00** | **1.00** | | **1.00** | **1.00** | **1.00** | | **0.86** | **0.86** | **0.86** |
| **macro avg** | **1.00** | **1.00** | **1.00** | | **1.00** | **1.00** | **1.00** | | **0.86** | **0.87** | **0.86** |
| **weighted avg** | **1.00** | **1.00** | **1.00** | | **1.00** | **1.00** | **1.00** | | **0.89** | **0.86** | **0.86** |
| **samples avg** | **1.00** | **1.00** | **1.00** | | **1.00** | **1.00** | **1.00** | | **0.86** | **0.86** | **0.86** |

**TABLE 5.** This table illustrates the effects of varying window sizes on the classification performance of the datasets. The figure presents the classification results across different window sizes, highlighting how the choice of window size influences the accuracy and overall performance of the model.

| No. | Window Size | Measure | PU | SA | IP |
|---|---|---|---|---|---|
| 1. | 5 | Kappa | 99.07 | 99.59 | 81.66 |
| 2. | | OA | 99.17 | 99.62 | 82.81 |
| 3. | | AA | 99.17 | 99.62 | 82.81 |
| 4. | 7 | Kappa | 99.86 | 99.92 | 80 |
| 5. | | OA | 99.88 | 99.93 | 81.25 |
| 6. | | AA | 99.88 | 99.93 | 81.25 |
| 7. | 9 | Kappa | 99.86 | 100 | 82.83 |
| 8. | | OA | 99.88 | 100 | 84.37 |
| 9. | | AA | 99.88 | 100 | 84.37 |
| 10. | 11 | Kappa | 99.80 | 100 | 93.33 |
| 11. | | OA | 99.82 | 100 | 93.75 |
| 12. | | AA | 99.82 | 100 | 93.75 |

### C. CLASSIFICATION RESULT

To assess the classification performance, we compared our proposed model to state-of-the-art machine learning (ML), 2D-CNN models, AD-GAN, Hybrid Spectral-Spatial Network (HybridSN), and Spectral Former(SF). The model was trained for 100 epochs using both real and synthetically generated data. The results demonstrate that our model achieves superior classification accuracy compared to the baselines. In the baseline model, SSRN gives better accuracy for all the dataset.

For the PU dataset, SSRN achieves an overall accuracy OA of 95.74%, while SB-GAN reaches 99.88%, showing a 4% improvement in OA. Furthermore, the AA for SSRN is 90.17%, compared to 99.88% for SB-GAN, reflecting a 9% increase.

For the SA dataset, SSRN's OA is 98.41%, whereas SB-GAN achieves 100%. The AA for SSRN is 95.69%, and for SB-GAN, it is 100%, resulting in a 1.5% increase in OA and a 4.5% increase in AA. For the SA dataset, the rare class contains a moderate number of samples compared to the IP and PU datasets. The data generation using cGAN performed effectively, resulting in a comparatively larger number of samples available for classifier training. Consequently, the OA, AA, and Kappa coefficient all achieve a value of 100%.

However, the OA of the Indian Pines dataset is comparatively lower. The OA of the SSRN is 94.99 % and SB-GAN has 84.37. The AA of the SSRN is 77.77 % but the AA of the SB-GAN is 84.37 %. This can be attributed to the fact that the Indian Pines dataset contains classes with a significantly limited number of samples. The SB-GAN focuses on rare classes to enable the cGAN to learn unbiased features. In Indian Pines, the smallest class has only 20 samples. Creating a balanced dataset with such a small number of samples is not sufficient for the cGAN to effectively learn the features.

The limited availability of real samples for cGAN training potentially hinders its performance, leading to the generation of less accurate synthetic samples. This, in turn, could explain the lower classification accuracy observed on the Indian Pines dataset. We can opt for multiple solutions to deal with such scenarios, such as transfer learning on a similar dataset and transferring the learned weights. Another potential solution is not using undersampling for the cGAN. In that case, we might achieve good OA, but we may not achieve good AA and class-wise accuracy for rare classes. Using this approach will cause the cGAN and 2D-CNN to adopt a biased tendency toward the majority classes.

Across all experiments, the results consistently show that the OA equals the AA, indicating a balanced class distribution in the dataset. This uniformity suggests that the classifier performs consistently across all classes, without bias towards any specific one. OA measures the proportion of correctly

**TABLE 6.** This table presents the classification results for each class of the PU dataset, comparing the performance of different methods. Additionally, the table includes the model training times for each method, providing a comprehensive view of both classification accuracy and computational efficiency across the various approaches.

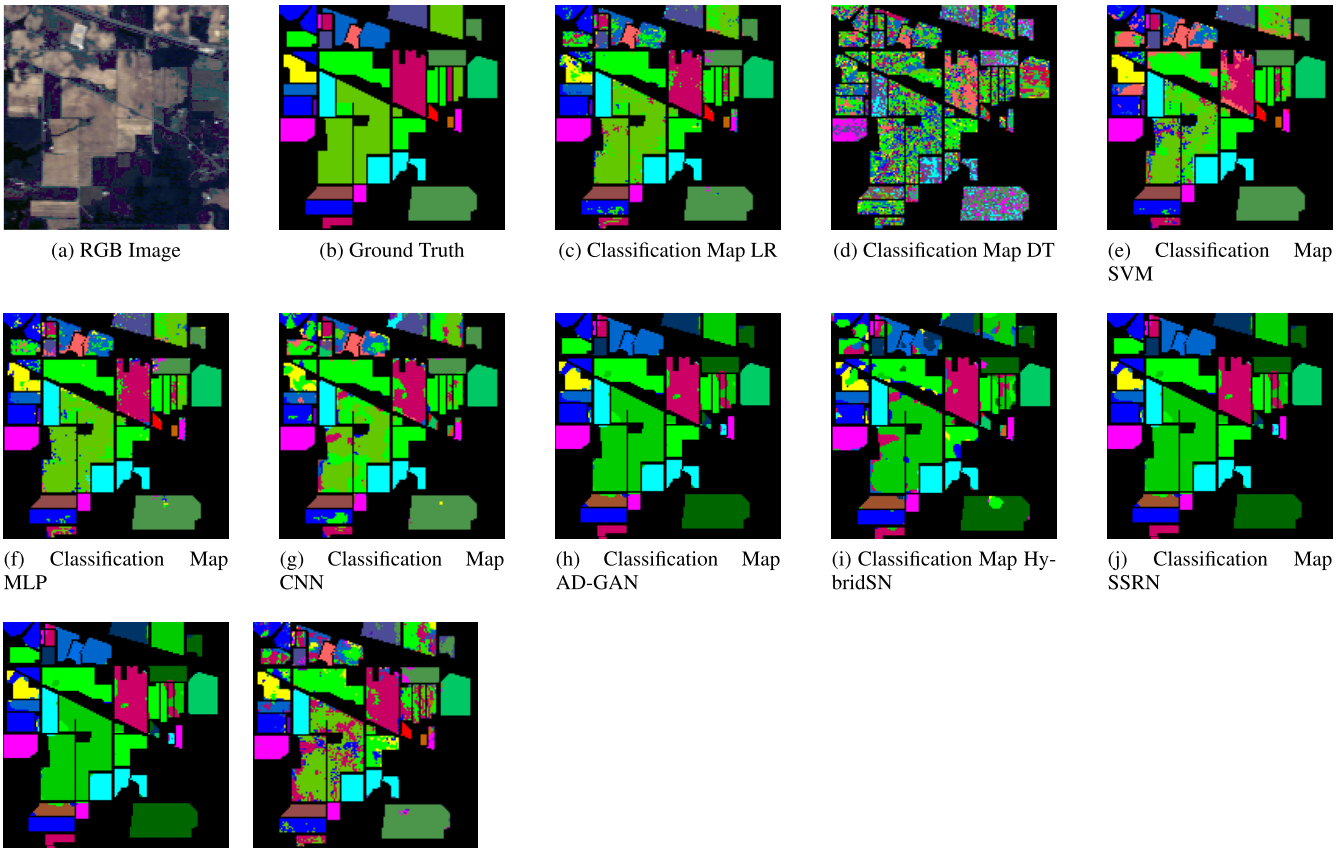
| No. | Class Name | LR | DT | SVM | MLP | CNN | AD-GAN | HybridSN | SSRN | SF | SB-GAN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | water | 0 | 32.46 | 100 | 91.29 | 91.21 | 91.70 | 94.18 | 97.54 | 95.11 | 99.47 |
| 2 | trees | 0.14 | 15.99 | 66.16 | 100 | 100 | 97.26 | 99.84 | 99.98 | 99.97 | 100 |
| 3 | asphalt | 2.31 | 7.49 | 35.19 | 89.10 | 88.83 | 98.50 | 87.15 | 94.89 | 66.98 | 100 |
| 4 | self blocking bricks | 46.11 | 94.36 | 67.69 | 82.23 | 81.72 | 94.32 | 84.70 | 93.40 | 82.45 | 100 |
| 5 | bituem | 56.36 | 94.58 | 100 | 100 | 100 | 100 | 100 | 99.77 | 97.37 | 100 |
| 6 | tiles | 2.41 | 56.96 | 100 | 31.98 | 31.05 | 100 | 97.24 | 95.84 | 92.56 | 100 |
| 7 | shadows | 0 | 0.11 | 35.76 | 98.28 | 98.28 | 99.84 | 91.41 | 86.94 | 53.37 | 100 |
| 8 | meadows | 0 | 63.75 | 30.19 | 86.37 | 86.65 | 97.36 | 71.98 | 84.66 | 64.41 | 99.47 |
| 9 | bare soil | 100 | 98.79 | 100 | 99.84 | 99.84 | 78.88 | 96.15 | 58.48 | 68.51 | 100 |
| | Kappa | 4.75 | 25.72 | 63.95 | 82.93 | 82.71 | 95.22 | 92.3 | 94.3 | 90.1 | 99.86 |
| | OA | 7.75 | 36.48 | 71.75 | 87.62 | 87.46 | 94.24 | 94.21 | 95.74 | 90.19 | 99.88 |
| | AA | 23.03 | 51.61 | 70.55 | 86.57 | 86.40 | 95.32 | 91.41 | 90.17 | 80.08 | 99.88 |
| | Train Time | 3.67 | 45 | 22.4 | 155 | 220.7 | 720 | 803 | 112 | 945 | 640 |

**TABLE 7.** This table presents the classification results for each class of the SA dataset, comparing the performance of different methods. Additionally, the table includes the model training times for each method, providing a comprehensive view of both classification accuracy and computational efficiency across the various approaches.

| No. | Class Name | LR | DT | SVM | MLP | CNN | AD-GAN | HybridSN | SSRN | SF | SB-GAN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Brocoli_green_weeds_1 | 8.94 | 90.90 | 100 | 92.24 | 90.61 | 100 | 100 | 99.94 | 99.79 | 100 |
| 2 | Brocoli_green_weeds_2 | 9.92 | 5.21 | 100 | 99.92 | 99.92 | 97.28 | 100 | 100 | 100 | 100 |
| 3 | Fallow | 0 | 0 | 0 | 0 | 0 | 99.49 | 99.23 | 100 | 100 | 100 |
| 4 | Fallow_rough_plow | 0 | 0.41 | 0 | 100 | 99.89 | 96.70 | 98.55 | 99.92 | 91.95 | 100 |
| 5 | Fallow_smooth | 0 | 95.2 | 0 | 95.94 | 91.46 | 9906 | 99.35 | 98.49 | 98.60 | 100 |
| 6 | Stubble | 0 | 95.09 | 99.89 | 100 | 100 | 99.92 | 99.94 | 100 | 100 | 100 |
| 7 | Celery | 0 | 0.72 | 100 | 99.88 | 99.28 | 100 | 100 | 99.88 | 99.18 | 100 |
| 8 | Grapes_untrained | 0 | 0 | 14.39 | 81.08 | 88.02 | 99.33 | 95.42 | 99.48 | 93.07 | 100 |
| 9 | Soil_vinyard_develop | 0 | 16.51 | 0 | 0.13 | 0.16 | 100 | 100 | 100 | 100 | 100 |
| 10 | Corn_senesced_green_weeds | 4.35 | 13.03 | 100 | 98.21 | 98.16 | 98.18 | 97.68 | 99.19 | 98.02 | 100 |
| 11 | Lettuce_romaine_4wk | 0 | 0 | 0 | 98.12 | 72.45 | 98.96 | 96.68 | 99.05 | 93.37 | 100 |
| 12 | Lettuce_romaine_5wk | 0 | 2.37 | 0 | 0 | 0 | 99.63 | 99.73 | 100 | 100 | 100 |
| 13 | Lettuce_romaine_6wk | 0 | 0 | 0 | 0 | 0 | 95.15 | 98.01 | 39.47 | 91.06 | 100 |
| 14 | Lettuce_romaine_7wk | 0 | 0 | 0 | 100 | 100 | 96.88 | 97.92 | 97.92 | 98.58 | 100 |
| 15 | Vinyard_untrained | 1 | 7.95 | 14.23 | 5.11 | 2.26 | 48.43 | 92.03 | 98.17 | 91.64 | 99.45 |
| 16 | Vinyard_vertical_trellis | 1.32 | 4.19 | 100 | 99.68 | 99.60 | 98.26 | 99.38 | 99.49 | 99.94 | 100 |
| | Kappa | 14.17 | 13.95 | 34.37 | 58.09 | 58.19 | 91.37 | 97.2 | 98.2 | 96.21 | 100 |
| | OA | 24.02 | 19.42 | 38.82 | 62.31 | 62.54 | 92.28 | 97.56 | 98.41 | 96.66 | 100 |
| | AA | 18.87 | 20.72 | 39.29 | 66.89 | 65.11 | 95.45 | 98.37 | 95.69 | 97.20 | 100 |
| | Train Time | 9 | 61.6 | 38.5 | 240.5 | 316 | 899 | 900 | 1326 | 1023 | 617 |

**TABLE 8.** This table presents the classification results for each class of the IP dataset, comparing the performance of different methods. Additionally, the table includes the model training times for each method, providing a comprehensive view of both classification accuracy and computational efficiency across the various approaches.

| No. | Class Name | LR | DT | SVM | MLP | CNN | AD-GAN | HybridSN | SSRN | SF | SB-GAN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Alfalfa | 90.62 | 31.25 | 93.75 | 78.12 | 84.37 | 86.95 | 6.66 | 9.60 | 4.54 | 100 |
| 2 | Corn-notill | 19.4 | 47.1 | 83 | 21.2 | 62.1 | 96.42 | 74.58 | 93.80 | 74.50 | 50.97 |
| 3 | Corn-mintill | 44.23 | 40.64 | 99.31 | 97.93 | 100 | 87.42 | 79.37 | 99.61 | 64.38 | 50.23 |
| 4 | Corn | 80.12 | 18.67 | 98.79 | 78.91 | 91.56 | 90.29 | 56.08 | 75.55 | 16 | 75.08 |
| 5 | Grass-pasture | 97.63 | 41.71 | 99.70 | 95.26 | 99.70 | 93.58 | 96.80 | 97.38 | 92.37 | 100 |
| 6 | Grass-trees | 81.40 | 78.47 | 99.60 | 96.28 | 99.41 | 98.49 | 98.87 | 98.84 | 95.23 | 100 |
| 7 | Grass-pasture-mowed | 55 | 78.47 | 90 | 75 | 85 | 100 | 70.37 | 20.41 | 3.70 | 100 |
| 8 | Hay-windrowed | 99.40 | 55.0 | 100 | 98.50 | 100 | 99.37 | 100 | 99.11 | 97.35 | 100 |
| 9 | Oats | 28.57 | 95.82 | 57.14 | 7.14 | 78.57 | 0 | 36.84 | 10.87 | 5.26 | 100 |
| 10 | Soybean-notill | 22.20 | 7.14 | 48.08 | 89.41 | 65.29 | 96.79 | 86.00 | 93.49 | 68.25 | 50.62 |
| 11 | Soybean-mintill | 99.88 | 48.53 | 81.73 | 80.91 | 86.32 | 99.67 | 83.16 | 97.94 | 83.49 | 50.54 |
| 12 | Soybean-clean | 13.73 | 60.67 | 73.73 | 45.06 | 88.91 | 99.32 | 69.21 | 90.76 | 35.70 | 75.23 |
| 13 | Wheat | 96.50 | 33.25 | 100 | 100 | 100 | 96.56 | 93.46 | 86.66 | 93.33 | 100 |
| 14 | Woods | 84.65 | 7.69 | 96.83 | 83.86 | 95.92 | 99.84 | 92.99 | 99.08 | 95.67 | 100 |
| 15 | Buildings-Grass-Trees-Drives | 27.78 | 79.23 | 97.41 | 62.59 | 99.81 | 99.44 | 76.47 | 95.09 | 54.22 | 100 |
| 16 | Stone-Steel-Towers | 92.30 | 44.44 | 98.47 | 89.23 | 98.46 | 100 | 74.44 | 76.13 | 73.86 | 100 |
| | Kappa | 58.13 | 49.85 | 84.14 | 72.07 | 84.44 | 96.63 | 81.40 | 93.80 | 72.91 | 84.37 |
| | OA | 64.87 | 55.79 | 86.04 | 75.20 | 86.27 | 96.94 | 83.66 | 94.99 | 76.60 | 84.37 |
| | AA | 64.59 | 46.56 | 88.59 | 74.96 | 89.34 | 90.26 | 74.70 | 77.77 | 59.86 | 84.37 |
| | Train Time in seconds | 4 | 19 | 13 | 63 | 71.75 | 799 | 821 | 923 | 833 | 61 |



(a) RGB Image (b) Ground Truth (c) Classification Map LR (d) Classification Map DT (e) Classification Map SVM (f) Classification Map MLP (g) Classification Map CNN (h) Classification Map AD-GAN (i) Classification Map HybridSN (j) Classification Map SSRN (k) Classification Map SpectralFormer (l) Classification Map SB-GAN

**FIGURE 4. Classification Map of IP Scene: (a) Shows the RGB image of PC using bands ( 29,20,11). (b) Shows the ground truth of the dataset. (c) Shows the classification map generated through LR classifier. (d) Shows the classification map generated through DT classifier. (e) Shows the classification map generated through SVM classifier. (f) Shows the classification map generated through MLP classifier. (g) Shows the classification map generated through AD-GAN. (h) Shows the classification map generated through HybridSN. (i) Shows the classification map generated through SSRN. (j) Shows the classification map generated through SpectralFormer. (k) Shows the classification map generated through SB-GAN classifier.**

classified instances across the entire dataset, while AA calculates the mean accuracy across all classes. The equality of OA and AA indicates the model's consistent performance across diverse classes, mitigating biases towards dominant classes. This balanced performance across categories is a desirable outcome, reflecting the model's robustness in handling different classes. We further calculated the precision, recall, and f1 scores of the classes and the results are given in Table 4.

### D. PARAMETER ANALYSIS

We further checked the model accuracy by different patch sizes, $5 \times 5$, $7 \times 7$, $9 \times 9$ and $11 \times 11$. We used $9 \times 9$ because $9 \times 9$ returns almost similar results as $11 \times 11$ with reduced size and efficient computation. We found that the proposed model, which returns quite good results is compared to other models. The model is compared based on $\kappa$, OA, and AA. The results for three datasets are shown in Table 5.

From the above table, we can see that PU and SA datasets are giving promising results on classification. We have got 99.96 % accuracy with SA dataset. The more we increase the patch size, the more accuracy is increased. The reason behind this is that a larger patch size gives more spatial information with a label. For further classwise accuracy, we have used patch size $9 \times 9$ because of its good performance and less system overhead.

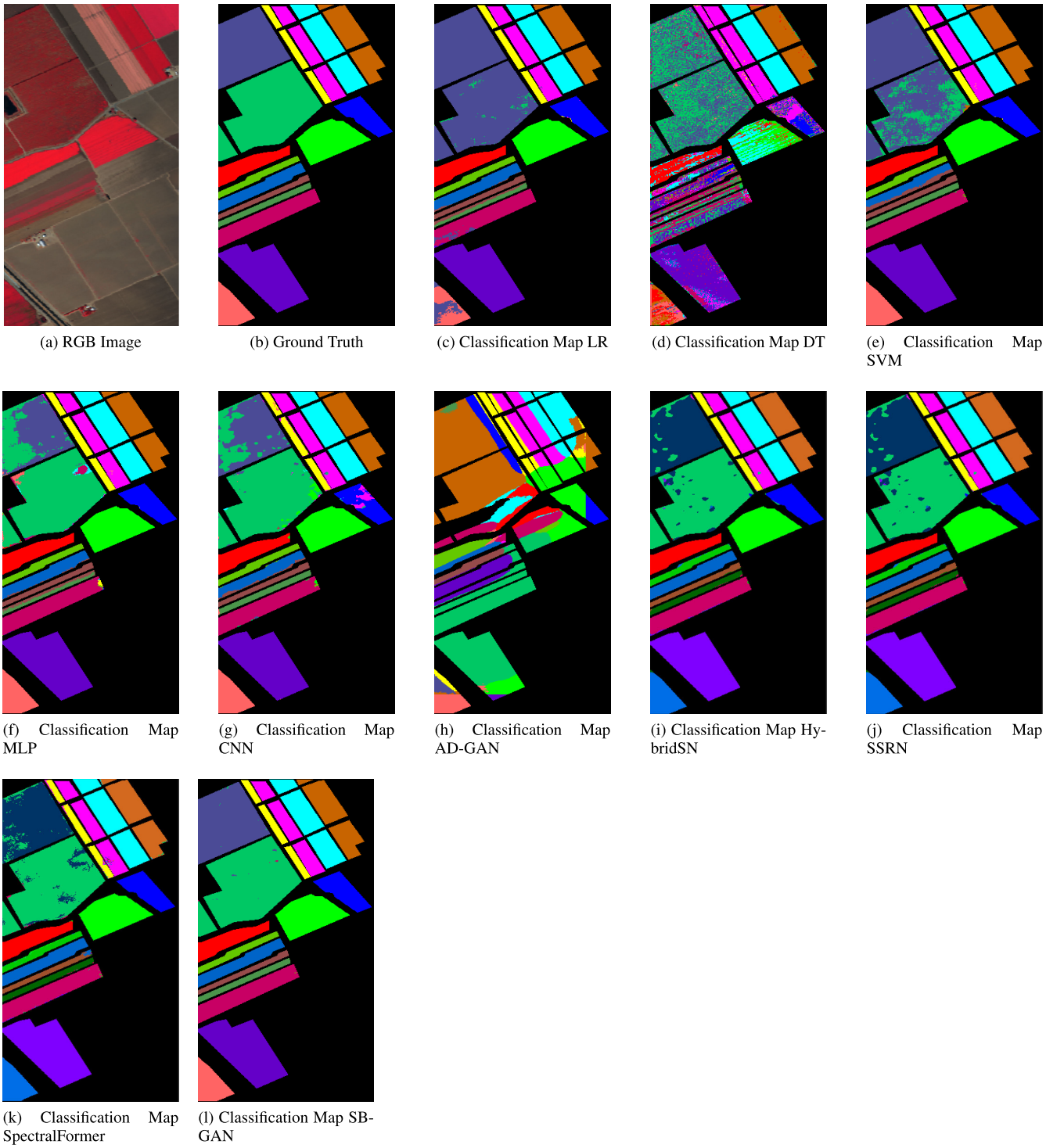
### E. ANALYSIS OF RUNNING TIME

To compare the run time complexity, we have computed the training time of all the models. We found that ML based models are efficient as compared to CNN, SB-GAN and SB-GAN based models but return less accuracy while deep learning based models especially cGAN model takes larger run time but have better classification performance. For the SB-GAN model, we have taken the train time of cGAN and the 2D-CNN.

Tables 6, 8, and 7 show the classification results for individual classes with the Kappa, OA, and AA measures for each dataset.

### F. VISUALIZATION

Inorder to have a clear idea of the outputs and the predicted labels, we take the complete HSI scene, pre-process it, and



(a) RGB Image (b) Ground Truth (c) Classification Map LR (d) Classification Map DT (e) Classification Map SVM

(f) Classification Map MLP (g) Classification Map CNN (h) Classification Map AD-GAN (i) Classification Map HybridSN (j) Classification Map SSRN

(k) Classification Map SpectralFormer (l) Classification Map SB-GAN

**FIGURE 5. Classification Map of SA Scene: (a) Shows the RGB image of PC using bands (60,34, 18). (b) Shows the ground truth of the dataset. (c) Shows the classification map generated through LR classifier. (d) Shows the classification map generated through DT classifier. (e) Shows the classification map generated through SVM classifier. (f) Shows the classification map generated through MLP classifier. (g) Shows the classification map generated through AD-GAN. (h) Shows the classification map generated through HybridSN. (i) Shows the classification map generated through SSRN. (j) Shows the classification map generated through SpectralFormer. (k) Shows the classification map generated through SB-GAN classifier.**

predict it through SB-GAN model. For better illustrative presentation, the RGB scene, the ground truth, and the classification map generated by reduced spectral features and classified through the decision tree are shown in Figures 3-5.

### G. COMPUTATIONAL COST

Training the cGAN model requires significant computational resources due to the iterative nature of adversarial training. The high training time can be attributed to the cGAN's complex loss computation and the need for simultaneous

**TABLE 9.** Table showing the performance metrics (Kappa, OA, and AA) for various experiments conducted as part of the ablation study. The experiments evaluate the impact of different configurations of Discriminator, Generator, and Classifier architectures using Conv2D and Conv2D-Transpose layers with ReLU and LeakyReLU activations across three datasets: PU, SA, and IP.

| Experiment | PU | | | SA | | | IP | | |
|---|---|---|---|---|---|---|---|---|---|
| | Kappa | OA | AA | Kappa | OA | AA | Kappa | OA | AA |
| **Discriminator** | | | | | | | | | |
| 1 Conv2D+ ReLU | 99.80 | 99.82 | 99.82 | 99.92 | 99.93 | 99.93 | 85.0 | 85.93 | 85.93 |
| 1 Conv2D+ LeakyReLU | 99.86 | 99.88 | 99.88 | 99.92 | 99.93 | 99.93 | 90.0 | 90.62 | 90.62 |
| 2 Conv2D+ ReLU | 99.86 | 99.88 | 99.88 | 100 | 100 | 100 | 85.0 | 85.93 | 85.93 |
| 2 Conv2D+ LeakyReLU | 99.86 | 99.88 | 99.88 | 100 | 100 | 100 | 84.37 | 84.37 | 84.37 |
| **Generator** | | | | | | | | | |
| 1 Conv2D-T+ ReLU | 99.80 | 99.82 | 99.82 | 100 | 100 | 100 | 85.0 | 85.93 | 85.93 |
| 1 Conv2D-T+ LeakyReLU | 99.80 | 99.82 | 99.82 | 100 | 100 | 100 | 95.0 | 95.31 | 95.31 |
| 2 Conv2D-T+ ReLU | 99.53 | 99.58 | 99.58 | 100 | 100 | 100 | 88.03 | 89.06 | 89.06 |
| 2 Conv2D-T+ LeakyReLU | 99.86 | 99.88 | 99.88 | 100 | 100 | 100 | 84.37 | 84.37 | 84.37 |
| **Classifier** | | | | | | | | | |
| 1 Conv2D+ ReLU | 99.86 | 99.88 | 99.88 | 99.96 | 99.90 | 99.90 | 81.67 | 82.81 | 82.81 |
| 1 Conv2D+ LeakyReLU | 99.73 | 99.76 | 99.76 | 99.96 | 99.96 | 99.96 | 80.02 | 81.25 | 81.25 |
| 2 Conv2D+ ReLU | 99.86 | 99.88 | 99.88 | 99.96 | 99.90 | 99.90 | 81.67 | 82.81 | 82.81 |
| 2 Conv2D+ LeakyReLU | 99.86 | 99.88 | 99.88 | 100 | 100 | 100 | 84.37 | 84.37 | 84.37 |

generator-discriminator updates. In contrast, SB-GAN is lightweight, efficient, and capable of learning complex features effectively. In SB-GAN, the cGAN model has a total of 610,238 trainable parameters, while the 2D-CNN model has 52,784 trainable parameters. Experiments were conducted on Google Colab using an NVIDIA Tesla T4 GPU with 12 GB VRAM and 16 GB of RAM. The cGAN model took approximately 120 ms per epoch, whereas the 2D-CNN required 100 ms per epoch. This reduction in training time is due to the down-sampling of data, as we focused on a lightweight, balanced dataset that uses only requisite number of samples from each class during the training of cGAN and 2D-CNN. Additionally, the use of the Adam optimizer further accelerated computation.

### H. ABLATION STUDY

The following table shows the ablation study of the model. We have tested the model by changing number of layers and activation functions. We iteratively checked the model performance on all the datasets with ReLU and LeakyReLU activation function. We further changed the number of layers for descriminator, generator, and classifier. We observe a common behavior among PU and SA datasets, that two layers with LeakyReLU shows better results in all cases. While for IP, ReLU gives better results both for single layer and two layers models. The results are shown in Table 9.

## V. CONCLUSION

Hyperspectral Imaging is crucial for land cover classification to provide detailed spectral information and capture a wider range of wavelengths. It enables more accurate identification and distinction of land cover types, such as forests, crops, and urban areas.

The scarcity of labeled samples in HSI classification generates a significant challenge. Copping the scarcity is critical for improving classification accuracy which is tackled through generative models. These models augment existing labeled data along with deep learning models to perform effectively in this context. These models facilitate the augmentation process through enhancing the performance of HSI classification models. However, the performance of deep learning models is often compromised by the inherent class imbalance present in land cover datasets. This imbalance arises when certain land cover types, such as urban areas or water bodies, are underrepresented as compared to forests or agricultural fields. The disproportionate representation of classes can lead to biased models that underperform in identifying minority classes. Therefore it is affecting the overall accuracy and reliability of the classification system.

Generative model is used to address the class imbalance problem by augmenting the underrepresented class through oversampling. Whereas, the cGANs are especially advantageous in this context, as they enable the generation of realistic synthetic data essential for enhancing model performance while maintaining generalization. Additionally, cGANs produce class-specific samples focusing on the necessary samples for minority classes. Latterly, the 2D-CNN model is used to classify the enlarged dataset which is the combination of real and augmented samples.

The results demonstrate that the proposed model has achieved better classification accuracy compared to the baselines. The model performs with an OA of up to 99% for the Salinas and Pavia University datasets, while 84% for the Indian Pines dataset. The Indian pines dataset has reflected lower accuracy due to the smaller dataset size and the lesser samples in minority classes. The limited number of real samples for cGAN training potentially hinders its performance. It leads to the generation of less accurate synthetic samples, which triggers the lower classification accuracy observed on the Indian Pines dataset.

The proposed SB-GAN model's indicate OA equals AA with improved accuracy due to the balanced class distribution. This uniformity suggests that the classifier performs consistently across all classes, without biasness. Additionally, increasing the patch size improves accuracy, as a larger patch size provides more spatial information

with each label. Machine learning-based models have higher efficiency as compared to CNN and SB-GAN models with lower accuracy. In contrast, deep learning-based models, especially the cGAN model, had better classification performance with longer runtime.

For future work, we propose extending our cGAN-based model to a multi-task learning framework that simultaneously performs HSI classification and reconstruction. Additionally, we aim to develop models capable of joint spectral and spatial classification, effectively exploiting both spectral and spatial features. Additionally, extending the approach to other imbalanced classification tasks and addressing lower performance on specific datasets through techniques such as semi-supervised learning or domain adaptation could provide valuable opportunities for further improvement. This dual-focus approach may enhance the model's robustness and accuracy. It will broaden its applicability to various real-world scenarios where detailed spectral-spatial information is crucial.

## CONFLICT OF INTEREST

The authors declare no conflict of interest.

## ACKNOWLEDGMENT

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R300), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia. The authors express sincere gratitude to Hosei University, Japan, for providing research facilities and technical support under Grant HU24K14859, which greatly supported this research.

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